

Question Number	Answer	Additional guidance	Mark
3(a)	A calculation in which: • calculation of volume of liposome (1) • expressed in standard form (1)	<u>Example of calculation:</u> 220893 $2 \times 10^5 / 2.2 \times 10^5 / 2.21 \times 10^5 \text{ (nm}^3\text{)}$ answer given to 3 sig fig ecf applied for mp2 for correct volume calculation using 75 as radius	(2)

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3(b)	An explanation that makes reference to the following points: Similarities • both are single-stranded (1) • both contain the bases A,U,C,G / both contain a base, phosphate group and ribose (sugar) / both contain phosphodiester bonds (1) • an exon on pre-mRNA contains same base sequence as the matching exon on active mRNA (1) Differences • pre-mRNA contains both introns and exons whereas active RNA only contains exons (1) • active mRNA may have {fewer / different order of} exons (than pre-mRNA) (1)	ACCEPT linear ACCEPT both do not contain thymine ACCEPT both contain RNA nucleotides / both contain uracil ACCEPT both contain exons ACCEPT pre-mRNA contains introns whereas active RNA does not ACCEPT pre-mRNA is {longer / has more bases} than active mRNA ACCEPT pre-mRNA has recognition site for spliceosome whereas active RNA does not ACCEPT active mRNA has a different base sequence to pre-mRNA	(4)